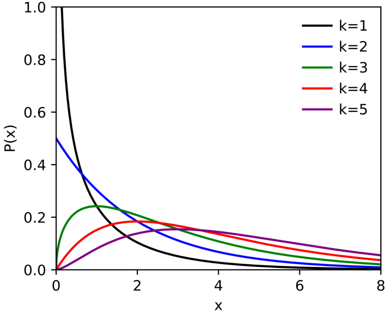
χ_k^2 DISTRIBUTION

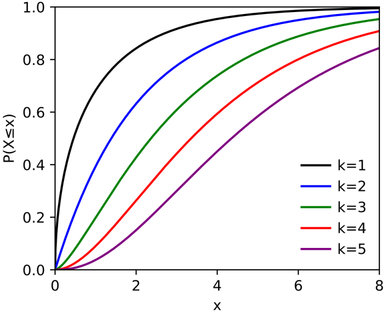
- $\chi_k^2 = \sum_{i=1,k} x_i^2 = x_1^2 + x_2^2 + \cdots + x_k^2$ with $x_i \approx N_{0,1}(x)$
- k=degrees of freedom

- DOMAIN OF χ_K^2 IS NONNEGATIVE
- IT IS NOT SYMMETRIC: SMALL VALUES TEND TO BE ASSOCIATED TO LARGER PROBABILITIES

density:



cumulative:



Sampling

inference based on population and samples

- population: large (infinite) collection of units on which a variable can be measured
- sample (with or without repetition): finite subset of units of the population
- random sample: a sample extracted randomly from the population

example:

- the mean of the age of a population of 1000 units is unknown
- a random sample of 100 units is extracted and its sample mean is computed

the sample mean is a random variable!

$\Rightarrow$ it must have a probability distribution (e.g. a mean and a variance)...

Inference

what is inference? two settings...

CONFIDENCE

- interval estimate:
  - I know a random variable distribution on a population
  - I observe the random variable on a RANDOM sample of the population

• THE FAMILY OF THE DISTRIBUTION
• A SUBSET OF PARAMETERS (MEAN, VAR,..)
• ...

can I estimate an interval in which a certain parameter (e.g. μ) of the distribution of the population may lie with high probability?

HYPOTHESIS TEST

- hypothesis rejection:
  - I know a random variable distribution on a population
  - I make an hypothesis on the value of a certain parameter of the distribution
  - I observe the random variable on a sample of the population

do the observations on the sample confim my hypothesis?

confidence interval for the mean

- a random variable X in a population is distributed as $N_{\mu,\sigma}(x)$ with σ known and μ unknown
- a random sample $S := \{x_1, x_2, \ldots, x_n\}$ is observed

N = NUMBER OF OBSERVATIONS IN THE RANDOM SAMPLE

- sample mean $\bar{x} = \frac{1}{n} \sum_{i=1,n} x_i$ is a random variable

SAMPLE ST. DEV.

it can be shown that $\bar{x} \approx N_{\mu, \frac{\sigma}{\sqrt{n}}}(\bar{x})$, i.e. $E(\bar{x}) = \mu$, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

$$\Rightarrow N_{\mu, \frac{\sigma}{\sqrt{n}}}\left(\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}\right) \approx N(Z)$$

we want determine an interval of values in which the unknown μ falls with probability $1 - \alpha$ (e.g. 95% if $\alpha = 0.05$)...

$1-\alpha \equiv$ CONFIDENCE LEVEL

$\alpha \equiv$ SIGNIFICANCE

exploit the fact that $\frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$ is a standard normal variable!

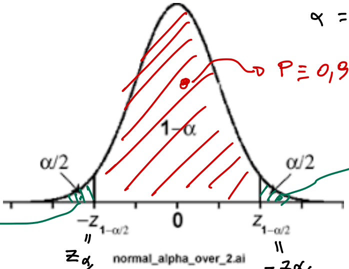
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

confidence interval for the mean

$1-\alpha = 0{,}95$

$\alpha = 0{,}05$

$\downarrow$ CENTERED AROUND THE MEAN



$P = 0{,}95$

$P = 0{,}025$

$0{,}025$

$-Z_{1-\alpha/2} = Z_{\alpha/2}$ $\qquad$ $Z_{1-\alpha/2} = -Z_{\alpha/2}$

$$P\{Z_{\frac{\alpha}{2}} \leq \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \leq Z_{1-\frac{\alpha}{2}}\} = 1-\alpha$$

$Z_{\frac{\alpha}{2}} \downarrow 0{,}025$ $\qquad$ $Z_{1-\frac{\alpha}{2}} \downarrow 0{,}975$ $\qquad$ $1-\alpha \downarrow 0{,}95$

$$P\{-\bar{x} + \frac{\sigma}{\sqrt{n}} Z_{\frac{\alpha}{2}} \leq -\mu \leq -\bar{x} + \frac{\sigma}{\sqrt{n}} Z_{1-\frac{\alpha}{2}}\} = 1-\alpha$$

$\downarrow$ CHANGE THE SIGNS OF INEQUALITIES

$$P\{\bar{x} - \frac{\sigma}{\sqrt{n}} Z_{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} - \frac{\sigma}{\sqrt{n}} Z_{\frac{\alpha}{2}}\} = 1-\alpha$$

$$P\{\bar{x} - \frac{\sigma}{\sqrt{n}} Z_{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} + \frac{\sigma}{\sqrt{n}} Z_{1-\frac{\alpha}{2}}\} = 1-\alpha$$

example:

the viral load of asymptomatic covid patients in L'Aquila is distibuted as a normal random variable with standard deviation 0.7 and unknown mean. The viral load is observed on a sample of 25 asymptomatic patients and the sample mean is 5.23 copies/ml (in Log10 scale). Define an interval of values $[a, b]$ in which the mean of the whole population lies with a 95% probability.

$$\text{DATA: } \begin{cases} \sigma = 0.7 \\ \bar{x} = 5.23 \\ n = 25 \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases} \qquad \text{UNKNOWNS: } \begin{cases} a \\ b \end{cases}$$

$$P\{5.23 - \frac{0.7}{\sqrt{25}} Z_{0.975} \leq \mu \leq 5.23 + \frac{0.7}{\sqrt{25}} Z_{0.975}\} = 0.95$$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077

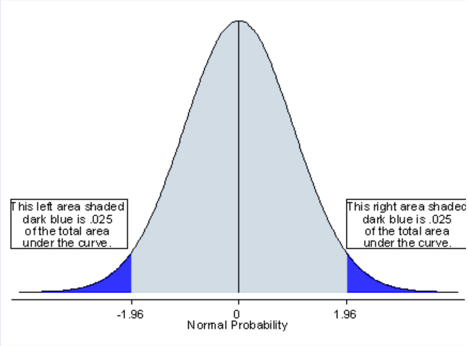
$$Z_{0.975} = 1.96, \quad \frac{0.7}{\sqrt{25}} = 0.14$$

$$a = 5.23 - 1.96 * 0.14 = 4.9556$$

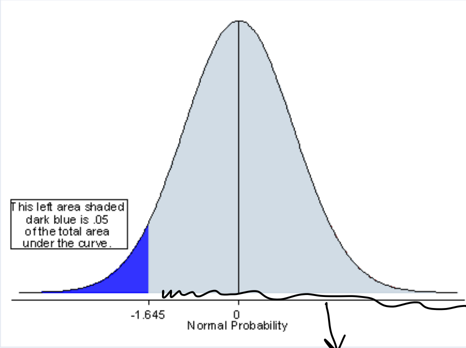
$$b = 5.23 + 1.96 * 0.14 = 5.5044$$

$$P\{4.9556 \leq \mu \leq 5.5044\} = 0.95$$

two-tailed confidence interval



one-tailed confidence interval



0,95 or 95%

what is the value a such the mean of the viral load is greater than or equal to a ($a \leq \mu$) with probability 95%?

LARGER α

$\Downarrow$

SMALLER INTERVAL

$\Downarrow$

MORE PRECISE INFO

$$P\{5.23 - \frac{0.7}{\sqrt{25}} Z_{0.95} \leq \mu\} = 0.95$$

$$Z_{0.95} = 1.645, \frac{0.7}{\sqrt{25}} = 0.14, \qquad a = 5.23 - 1.645 * 0.14 = 4.9997$$

$$P\{4.9997 \leq \mu\} = 0.95$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA